

# New Approach of *Beauveria bassiana* to Control the Red Palm Weevil (Coleoptera: Curculionidae) by Trapping Technique

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J. Econ. Entomol. 1–8 (2015); DOI: 10.1093/jee/tou055

**ABSTRACT** This work is the first study to investigate the efficacy of the commercial formulation of *Beauveria bassiana* (Broadband) to control adults of red palm weevil (*Rhynchophorus ferrugineus* (Olivier)). This fungus could be applied as one of the biological tactics in controlling red palm weevil. Bioassay experiments for medium lethal concentrate and medium time to cause death of 50% of red palm weevil adults were carried out. The result showed that the LC<sub>50</sub> of *B. bassiana* (Broadband) was  $2.19 \times 10^7$  and  $2.76 \times 10^6$  spores/ml at 9 and 23 d of treatment, respectively. The LT<sub>50</sub> was 13.95 and 4.15 d for concentration of  $1 \times 10^7$  and  $1 \times 10^8$  spores/ml, respectively, whereas  $1 \times 10^9$  spores/ml caused 100% mortality after 24 h. Additionally, a red palm weevil pheromone trap was designed to attract the adults to be contaminated with spores of Broadband, which was applied to the sackcloth fabric that coated the internal surfaces of the bucket trap. The mating behavior was studied to determine direct and indirect infection of the spores from male to female and vice versa. The results showed a high efficacy of Broadband suspension at  $1 \times 10^9$  spores/ml; 40 ml of suspension at this concentration treated to cloth in a trap caused death of contaminated adults with *B. bassiana* spores directly and indirectly. The 100% mortality was obtained even after 13 d of traps treatment with 40 ml of the suspension at  $1 \times 10^9$  spores/ml.

**KEY WORDS** red palm weevil, *Beauveria bassiana*, biological control, LC<sub>50</sub>, LT<sub>50</sub>

## Introduction

The red palm weevil (*Rhynchophorus ferrugineus* (Olivier)) was discovered in the Gulf Cooperation Council (GCC) countries in mid-1985 and in Saudi Arabia in early 1987 (Al-Abdulmohseen 1988). It is the most destructive pest of date palms in the GCC countries; therefore, the governments of the GCC invest and spend a substantial part of their domestic growth (DG) on controlling this pest. Trapping is considered the most effective for controlling palm weevils when aggregation pheromone is used in combination with kairomone. In addition, curative and preventive spray and injection of insecticides in palm trees is widely used to control the red palm weevil. The problem of invasive red palm weevil is not only in the Middle East only; it has been reported in Europe, North Africa, the United States, and the Caribbean Islands ([www.redpalmweevil.com](http://www.redpalmweevil.com)). In 1986, the USDA intercepted nearly 50,000 alien organisms at ports of entry (DeBach and Rosen 1991). Indeed, translocation of living organisms is one of the most serious ecological changes caused by humans.

The observations of the effects of entomopathogenic fungi date back thousands of years, when mycosis was observed in honeybees and silkworms (Castrillo et al. 2005). Entomopathogenic fungi as biological control

agents are viewed as promising microbial agents to control insect pests and are becoming an interesting alternative to chemical control (Shah and Pell 2003, Maniania et al. 2008, Vega et al. 2008).

More than 700 species of fungi from about 90 genera are known as insect pathogens (Charnley 1989, Inglis et al. 2001). The entomopathogenic fungi being used most widely to control insects are *Beauveria bassiana* and *Metarhizium anisopliae* (Langewald et al. 1997, Wraight et al. 2000, Wraight and Ramos 2002, Feng et al. 2004a,b, Pu et al. 2005, Meyers et al. 2013). For over 120 years, *B. bassiana*, *M. anisopliae*, and *B. brongniartii* have been used to control pest insects (Zimmermann 2007). These fungi have been developed as commercial biopesticides, for example, all stages of red oak borer, *Enaphalodes rufulus*, were susceptible to a natural isolate of (ARSEF 7404) or a commercial *B. bassiana* product, BotaniGard (Meyers et al. 2013), and (Broadband) supplied by Becker Underwood, Inc. using the virulent isolate of *B. bassiana*.

Zhang et al. (2011) reported the pathogenicity of *B. bassiana* on the larvae of the red turpentine beetle, *Dendroctonus valens*, such that the highest virulent isolate was Bb1801, which killed 100% of the treated larvae and had a median lethal time (LT<sub>50</sub>) of 4.60 d at a concentration of  $1 \times 10^7$  conidia/ml. Song and Fong (2011) found the conidial surface properties and pathogenicity (conidial hydrophobicity rates [Hr], zeta potentials [Pz], and subtilisin-like protease [Pr1] activities [Ap]) accounted for 89% to the mortality variation

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( $r=0.89$ ) when they studied the pathogenicity of 21 *B. bassiana* isolates from heterologous host insects against the brown plant hopper, *Nilaparvata lugens*.

This work was conducted to investigate the efficacy of the commercial formulation of *Beauveria bassiana* (Broadband) to control the red palm weevil.

### Material and Methods

**Test Insects.** The adults and larvae of red palm weevil were obtained from the laboratory of Date Palm Research Center of Excellence (DPRCE), King Faisal University, Saudi Arabia. Pineapple was used as source food media, because it is attractive to the females to lay eggs according to Abo-El-Saad, M. at DPRCE (personal communication); all larvae were reared on pineapple in cages (45 by 38 by 38 cm<sup>3</sup>) made of transparent polyethylene with wire mesh on the top placed under lab conditions (25–30°C) while the adults were reared on fresh palm frond and dates.

**The Entomopathogenic Fungus.** Becker Underwood, Inc. (West Sussex, UK.) supplied the formulation of *B. bassiana* (Broadband).

**Initial Experiments to Prove the Pathogenicity of Fungi.** Ten red palm weevil adults were surface sterilized with cotton wool soaked with ethanol (70%), prior to each individual adult being sprayed with 1 ml of a suspension consisting of spores of *B. bassiana* ( $1 \times 10^9$  spores/ml). After treatment, insects were kept in glass dishes covered with mesh cloth. The treated red palm weevil was inspected every 24 h and considered dead if it was immobile. The dead red palm weevil were put into sterile petri dishes with high humidity and incubated at  $25 \pm 1^\circ\text{C}$  in order to retrieve fungal spores.

**Conidial Suspensions.** Culture of isolated *B. bassiana* from red palm weevil was maintained on potato dextrose agar (Difco Laboratories, Detroit, MI). The fungi were incubated at  $26 \pm 1^\circ\text{C}$  in complete darkness. Spores were harvested by flooding each plate with 10 ml of 0.05% aqueous (v/v) Tween 80 (Sigma-Aldrich, St Louis, MO) as a wetting agent in sterile distilled water. The conidia were dislodged into the suspension with a glass rod. The suspension was filtered through a double layer of sterile cheesecloth (mesh: 36 by 36 threads/cm<sup>2</sup>) to remove clumps of mycelia. The spore suspension of  $1 \times 10^9$  spores/ml was adjusted by using Neubauer haemocytometer (0.0625 m<sup>2</sup>; Fuchs-Rosenthal Merck Euro Lab, Darmstadt, Germany). Viability of the spores was determined before each assay by examining 100 spores under a compound microscope  $\times 400$  magnification at five separated areas, after they were incubated for 24 h on potato dextrose agar at  $25 \pm 2^\circ\text{C}$  in complete darkness. If the germ tube was more than half the diameter of the spore, it was considered germinated.

**Bioassay.** The basic measure of virulence generated in this study was the median lethal concentration (LC<sub>50</sub>) expressed as viable conidia/ml applied to adult red palm weevil. The bioassay experiments were arranged as a completely randomized design with five

treatments (four concentrations + control) and three replicates.

**Contact Application of Fungi.** Two separated experiments were conducted; the first one with spore suspensions of *B. bassiana* of the formulation Broadband, the second one was with laboratory-prepared spore suspension from the culture of the isolated fungi from the treated red palm weevil. Adult red palm weevil were sprayed by using a spray chamber, which was fitted with an air-atomizing nozzle that was connected to a regulator valve providing a constant airflow of 10 liter/min. The target surface in the spray chamber was a rotating turntable (30 rpm). The bioassay treatments with four different spore concentrations ( $1 \times 10^9$ ,  $1 \times 10^8$ ,  $1 \times 10^7$ , and  $1 \times 10^6$  spores/ml) were serially diluted from the primary suspension of  $1 \times 10^9$  spores/ml, and water was used as control. Each replicate was composed of 10 adults placed in one cage. The sprayer was calibrated with 30 ml of each concentration, and was used for three replicates (30 adults). Therefore, each individual adult on filter paper placed in petri dish was sprayed directly with 1 ml of the appropriate concentration. After 15 min, the treated adults were transferred to their cages. To prevent cross contamination in the laboratory, spraying was carried out in isolated room under a fume cupboard. The control treatment was applied first, and then the other four treatments were started from low concentration to high. All suspensions were refrigerated (ca. 4°C) prior to use and were used within a few hours of preparation. The conditions in the laboratory were adjusted to a photoperiod of 15:9 (L:D) h and 75–80% relative humidity, with temperatures between 25–30°C.

**Records of Insect Mortality.** Each day after treatment for 30 d, the number of dead and survival of adults were recorded. Lethal concentrations (LC<sub>50</sub>) of the fungi based on four conidial concentrations were calculated, and lethal time (LT<sub>50</sub>) for each was calculated from the survival curve. The dead adults were moved into sterile petri dishes with high humidity ( $75 \pm 5\%$ ) and incubated at  $25 \pm 1^\circ\text{C}$  and the mycelia and spores were retrieved from the dead ones.

**Initial Experiments to Prove the Efficiency of Transferring the Infection From Treated Females to Untreated Adults.** This experiment was carried out to study the possibility of transferring the conidia of fungi from female to male and vice versa mainly through the mating behavior of the red palm weevil, and the pathogenic efficacy of the transferred conidia.

The initial experiment was performed in a laboratory; 15 males and 15 females were treated separately with fungi by keeping each five adults in contact with 5 ml of *B. bassiana* suspension at  $1 \times 10^9$  spores/ml in petri dish for 5 min. Each replicate of five treated males were transferred into a cage (30 by 30 by 20 cm<sup>3</sup>) and kept with five untreated females and vice versa. The cages were covered with muslin mesh on the top and were placed in the laboratory. The mortality of adults was recorded daily and for 30 d from the initial treatment.

**The Efficiency of Trapping System on Attraction, Infection, and Reinfection of Red Palm Weevil.** The red palm weevil trap was designed to attract and contaminate red palm weevil with conidia of *B. bassiana* from the commercial product (Broadband). The transfer of the conidia from the infected (directly exposed) adult to another uninfected adult was through the mating behavior. The trap was made of a small bucket (25 cm in height, with a circular bottom of 15 cm in diameter). Four symmetric holes (4 by 4 cm<sup>2</sup>) were made on the perimeter body of the bucket; the external and internal surfaces of the perimeter body of the bucket were coated with rough sackcloth fabric. The trap was supplied with food bait that was pieces of a palm frond and date fruits that were kept in the bottom of the bucket. The lure of Ferrugineol pheromone (ChemTica International, S.A. Costa Rica) was hanged on the top of bucket through the bucket's lead. Those combinations of attractants were already proven in attracting and trapping both female and male red palm weevil (Abbass and Al- Nasser 2012, Abbass and Mesallam 2013). The layer of sackcloth that covered the internal surface of the bucket was soaked with 40 ml of Broadband suspension at 1 × 10<sup>9</sup> spores/ml.

Two experiments with four trap replicates each were carried out separately, one with female and the other with male red palm weevil. Each replicate of five unisex adults was kept in an enclosed cage (60 by 50 by 40 cm<sup>3</sup>) with one infected trap with 40 ml of 1 × 10<sup>9</sup> spores/ml of *B. bassiana* conidia (Broadband). Thus, the trapping and infection experiment was carried out separately on 20 females and 20 males. The behavior of the insects toward the trap was monitored by camera, with the data of duration of time required for each insect to enter the trap and the time spent in it before it left the trap being daily recorded for 15 d. The time required for the last insect to enter the trap and the time for the last insect to leave the trap were taken into consideration. The infected males by direct exposure of each replicate were transferred into a box (30 by 30 by 20 cm<sup>3</sup>) and kept with five clean untreated females, and vice versa. The boxes were covered with muslin mesh on the top and placed in the laboratory. After 24 h, the males were separated from the females and each kept with food consisting of small pieces of fresh palm leaves and date diets. The mortality of the insects was recorded daily and for 30 d from the initial treatment.

**The Persistence of Trapping and Infection Efficiency.** This study was carried out by repeating the experiment of the trapping system for infection and reinfection daily, and for 14 d starting from the day of treating the internal coated layer of rough sackcloth fabric with suspension of 40 ml of 1 × 10<sup>9</sup> spores/ml of *B. bassiana* conidia (Broadband).

**Statistical Analysis.** Cumulative mortality at the end of the experiment was analyzed by ANOVA, The effect of fungi were compared by Tukey test to determine significant differences between the treatments using SAS 9.3 (2007). Time for 50% cumulative mortality LT<sub>50</sub> and concentration for 50% mortality LC<sub>50</sub> were determined using the probit analysis program (LDP Line).

**Results**

The result of preliminary tests showed that fungus (Broadband) supplied by Becker Underwood, Inc. was capable of infecting red palm weevil. Mycelia and spores retrieved from dead insects were identified using morphological characteristics as the fungal species that had originally been applied to the insects.

**Contact Application of Fungi.** The ready-made product of *Beauveria bassiana* (Broadband) was highly effective on red palm weevil. The insects were treated with 1 ml by sprayer and inspected every 24 h for 30 d of the treatment; the number of dead or survival of adults was counted. At a concentration of 1 × 10<sup>8</sup> spores/ml, the first death was recorded after 4 d of the application, while at concentrations of 1 × 10<sup>7</sup> and 1 × 10<sup>6</sup> spores/ml, the dead insects were recorded after 6 and 9 d, respectively. The highest cumulative mortality recorded at 1 × 10<sup>8</sup> was 100% after 16 d of treatment, whereas with 1 × 10<sup>7</sup> the highest mortality was 75% after 23 d. Therefore, two values of LC<sub>50</sub> were estimated, the first form data recorded for 9 d of treatments and the other for 23 d of treatment. The dead insects were put into sterile Petri dishes with high humidity and incubated at 25 ± 1°C, and the mycelia and spores were retrieved from the dead insects. The results in Table 1 show the LC<sub>50</sub> at 9 and 23 d after treatment were 2.19 × 10<sup>7</sup> and 2.76 × 10<sup>6</sup> and there was a significant difference between the toxicity of the product at those two intervals. These results are logical because more dead insects were recorded after 9 d of treatment, and the LC<sub>50</sub> in this test represented the cumulative toxicity of the fungi.

However, the probit analysis of the contact toxicity test of the laboratory-prepared suspension of spore isolated from treated red palm weevil showed that no significant regression could be estimated from the recorded data, because only 20% of treated insects with the isolated *B. bassiana* at 1 × 10<sup>9</sup> spores/ml were killed after 9 d of treatment, and 60% was the highest cumulative mortality after 3 wk of treatment, whereas the Broadband formula at same concentration caused 100% mortality after 24 h of the treatment.

The LT<sub>50</sub> values of cumulative mortality with time of Broadband<sup>TM</sup> concentrations at 1 × 10<sup>7</sup> and 1 × 10<sup>8</sup> spores/ml were estimated using probit analyses. The results represented that the Broadband formula of *B. bassiana* killed 50% of the adults of red palm weevil within 13.95 and 4.15 d at the 1 × 10<sup>7</sup> and 1 × 10<sup>8</sup> spores/ml, respectively (Table 2). The periods to kill red palm weevil are similar to the results of

**Table 1. Median lethal concentrations of Broadband *B. bassiana***

| LC <sub>50</sub> | Conc. spore/ml         | Lower limit           | Upper limit            | Slope ± SE   | χ <sup>2</sup> | r      |
|------------------|------------------------|-----------------------|------------------------|--------------|----------------|--------|
| 9 d              | 2.19 × 10 <sup>7</sup> | 7.9 × 10 <sup>6</sup> | 5.49 × 10 <sup>7</sup> | 0.932 ± 0.18 | 3.4456         | 0.9887 |
| 23 d             | 2.76 × 10 <sup>6</sup> | 4.1 × 10 <sup>5</sup> | 8 × 10 <sup>6</sup>    | 0.864 ± 0.23 | 0.808          | 0.9883 |

previous study by Bateman (1997) that insect field mortality caused by fungal pathogens rarely occurs earlier than 6 d after application. However, at highest concentration of  $1 \times 10^9$  spores/ml, the mortality was 100% after 24 h and that could be due to the amount of spores attached to the surface of weevil and to the unknown inert and synergist material in the Broadband formula, which may play role in synergizing and speeding the infection of the fungus; therefore, the  $LT_{50}$  value could not be calculated statistically.

Cumulative mortality at the end of the experiment was analyzed by using Tukey test. Figure 1 illustrated

the mean daily cumulative mortality (MDCM) of different treatments for both the Broadband formula and the isolated fungi. A nonsignificant difference was found only between the two highest concentrations of the Broadband formula. With the isolated *B. bassiana*, the MDCM at  $1 \times 10^9$  spores/ml was 32% and there were no significant differences between treatments. These results reflect that the isolated fungi were not as effective in killing the red palm weevil as Braodband.

**The Efficacy of Transferring Spores of *B. bassiana* from Treated to Untreated Adults.** The preliminary tests conducted in laboratory showed that the pathogenic efficacy of direct exposure with 1 ml/ insect of Broadband™ *B. bassiana* at  $1 \times 10^9$  spores/ml on red palm weevil was high and all directly exposed males or females were killed after 24 h of exposure. Furthermore, the directly infected (exposed) adults transferred the spores to clean adults of both males and females and all indirectly infected insects were killed. These initial results encouraged us to develop the red palm weevil trap of infection and reinfection system with Broadband.

Table 2. Median cumulative lethal time of Broadband *B. bassiana*

| Concentration   | CLT <sub>50</sub><br>(d) | Lower<br>limit | Upper<br>limit | Slope ± SE     | χ <sup>2</sup> | r      |
|-----------------|--------------------------|----------------|----------------|----------------|----------------|--------|
| $1 \times 10^7$ | 13.95                    | 10.9702        | 18.9705        | 2.5199 ± 0.669 | 3.9197         | 0.9579 |
| $1 \times 10^8$ | 4.1597                   | 3.4746         | 6.0098         | 2.389 ± 0.5180 | 3.4746         | 0.9769 |

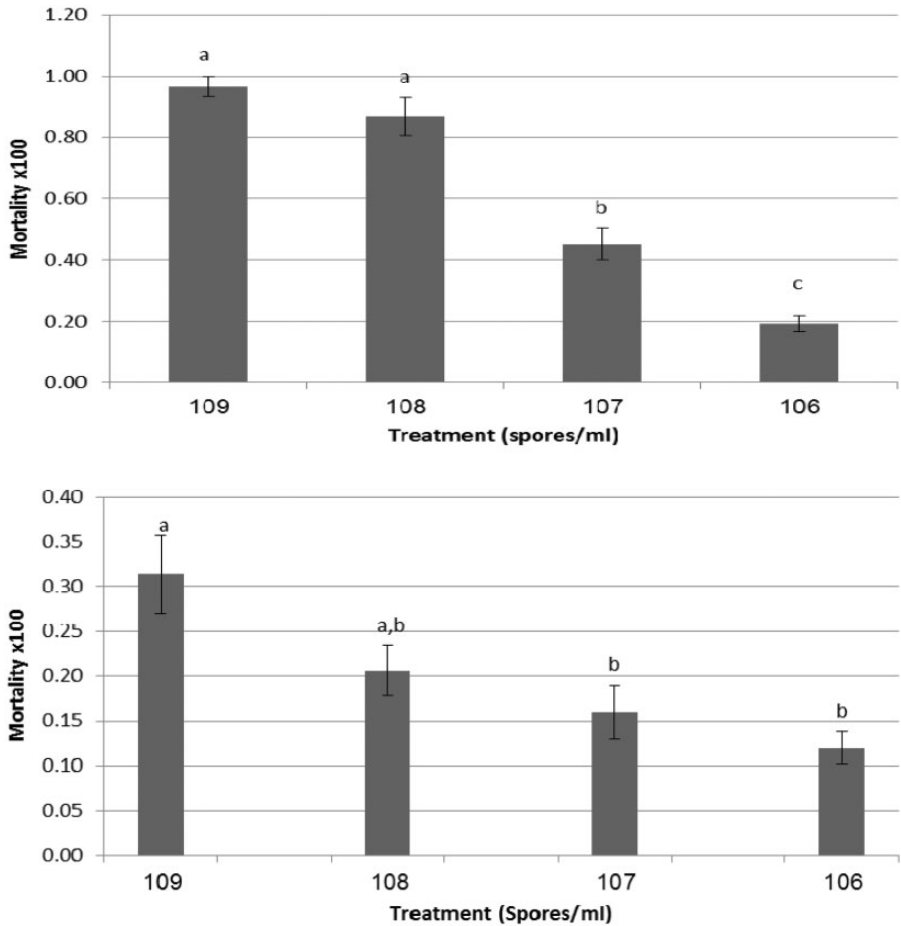


Fig. 1. The mean (±SE) percent mortality ( $n = 30$  adults and 3 replicates) of red palm weevil 25 d after contact treatment for Broadband formula and isolated *B. bassiana* from infected red palm weevil at different concentrations. Mean ± SE values with the same letter along the bars of different concentrations are not significantly different ( $P > 0.05$  Tukey test).

**The Persistence of Trapping, Attraction, and Infection to Red Palm Weevils with *B. bassiana*.**

The efficiency of attraction and infection was considered by recording the duration of time that was taken until the last insect of the 25 in the five replicates that entered the trap. In addition, the time that the trapped insects stayed in the traps before they left was also recorded (Table 3).

Females of red palm weevil entered the trap within 6 h in the first 3 d of the trap's age, but when the tested traps varied from 4 to 14 d old, the insects entered the traps within 8 h. The insects, after trapping, spent sometime in the trap before leaving. The length of time they stayed in the trap was variable according to the age of the tested trap. Females left the trap within 4 h when the trap's age was between 1–9 d old. After the trap was 10–14 d old, the females left the trap within 6 h of the last insect entering the trap.

In the male's case, the pheromone and kairomone in the trap were more attractive to males than females in the first 3 d of the trap's age. The insects entered the traps within 2 h, and then the entry time gradually increased until reaching 12 h when the traps were 11 d old. Afterwards, the male's attraction to the traps decreased, and at 12 and 14 d of the trap's age, the males left the traps within 24 and 48 h, respectively. This could be due to the reduction in the concentration of the pheromone within that time, and this reduction effected the attraction behavior of the males more than the females although the pheromone (Ferrugineol) used was an aggregation pheromone.

Moreover, the males stayed for some time in the trap before they left, and the duration of time varied according to the age of the traps. In the first 9 d of the trap's age, the males left the traps within 6 h, but when traps were more than 9 d old until the end of the experiment (14 d), the males left the trap within 10 h. This result shows that males were more attracted to the trap in the first week of the trapping age and spent more time than females in the trap before leaving. However, the results showed that the adults spent long enough in the trap to acquire a lethal infection.

**The Efficacy of Direct Infection.** The experiments were carried out with two separated tests (one with females and the other with male) and were repeated daily for 13 d on the same trap with each replicate to study the persistence of trapping efficacy in attraction, infection, and reinfection. The daily number of dead insects and the period required to reach 100% mortality with each daily test of the trap duration (13 d) were recorded. The results graphically were illustrated in (Fig. 2). The mortality of directly infected insects from 1-d-old traps was 100% after 24-h exposure for both males and females, and this result was sustained for the three following days of trap's age. However, when the trap's age was between 4 to 9 d old, the maximum cumulative mortality (100%) caused by Broadband to the females and also to males was observed after 2 d. But when the trap age was from 10 to 13 d, the cumulative mortality of 100% was recorded after 3–4 d.

**Table 3. The effect of trap age after application on time (h) required to enter<sup>a</sup> and to leave<sup>b</sup> the trap for both female and male red palm weevil**

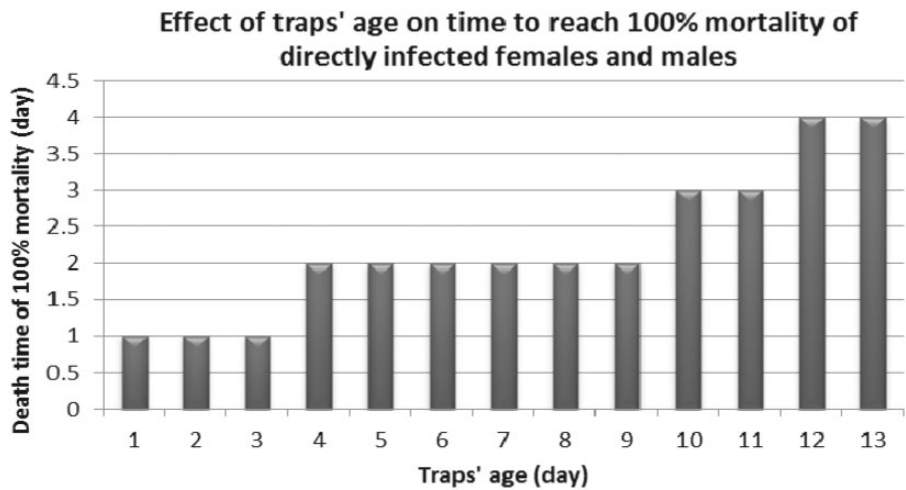
| Females of red palm weevil |                        |                       | Males of red palm weevil |                       |
|----------------------------|------------------------|-----------------------|--------------------------|-----------------------|
| Trap age (d)               | Time to enter the trap | Time stay in the trap | Time to enter the trap   | Time stay in the trap |
| 1                          | 6                      | 4                     | 2                        | 6                     |
| 2                          | 6                      | 4                     | 2                        | 6                     |
| 3                          | 6                      | 4                     | 2                        | 6                     |
| 4                          | 8                      | 4                     | 6                        | 6                     |
| 5                          | 8                      | 4                     | 6                        | 6                     |
| 6                          | 8                      | 4                     | 8                        | 6                     |
| 7                          | 8                      | 4                     | 8                        | 6                     |
| 8                          | 8                      | 4                     | 10                       | 6                     |
| 9                          | 8                      | 4                     | 10                       | 6                     |
| 10                         | 8                      | 6                     | 12                       | 10                    |
| 11                         | 8                      | 6                     | 12                       | 10                    |
| 12                         | 8                      | 6                     | 24                       | 10                    |
| 13                         | 8                      | 6                     | 24                       | 10                    |
| 14                         | 8                      | 6                     | 48                       | 10                    |

<sup>a</sup> Time required for the last insect to enter the trap.  
<sup>b</sup> Time required for the last insect to leave the trap.

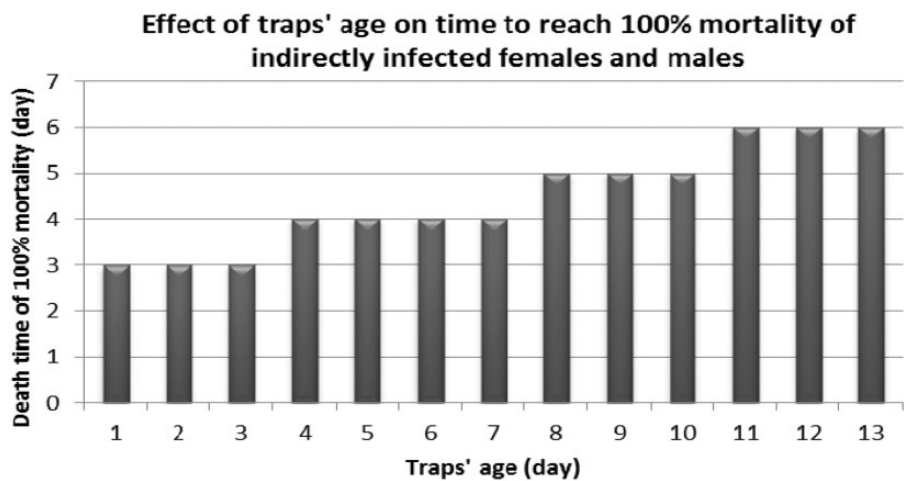
**The Efficacy of Indirect Infection.** Moreover, the infected males of each replicate were transferred into a box (30 by 30 by 20 cm<sup>3</sup>) and kept with five untreated females, and the infected females were kept with untreated males; then males and females were separated after 24 h. The mating behavior of the red palm weevil resulted in transferring the spores from infected insects to the uninfected insects. The pathogenic efficacy of *B. bassiana* indirectly infecting males and females is illustrated in Figure 3. The efficacy of indirect infection was high. But, the period required to obtain the maximum mortality of 100% was extended to compare the directly infected females or males at all intervals of trap's age. However, the time to death of indirectly infected males or females from mating with the infected insects ranged from 3 to 6 d after mating.

**Discussion**

This study demonstrates that the formulation of *B. bassiana* (Broadband) and the isolates used are capable of causing infection and mortality of the red palm weevil *R. ferrugineus* via contact. The formulation of *B. bassiana* (Broadband) at the same concentration caused more mortality than *B. bassiana* isolated from adults of the red palm weevil. The first death caused by the studied concentration at  $1 \times 10^8$ ,  $1 \times 10^7$ , and  $1 \times 10^6$  spores/ml of *B. bassiana* (Broadband) were observed at 4, 6, and 9 d after treatment, respectively. These times are similar to the results of the previous study by Bateman (1997) claiming that insect field mortality caused by pathogen rarely occurs earlier than 6 d after application, except at the highest concentration of  $1 \times 10^9$  spores/ml where 100% mortality was observed after 1 d of treatment. The formulation of *B. bassiana* (Broadband) took less time to infect and kill the insects, and this faster action was also recorded with all concentrations that were used in experiment. The inert material and synergist were unknown in the



**Fig. 2.** The persistence of *B. bassiana* (Broadband) trap in directly infecting and killing the female and male red palm weevils.



**Fig. 3.** The persistence of *B. bassiana* (Broadband) trap in indirectly infecting and killing the male and female red palm weevils from mating with the infected adults.

Broadband formulation, but they enhanced the speed of the infection of the fungi. Inglis et al. (1996) found the spores formulated in oil on lettuce discs were more effective against grasshopper nymphs than those formulated in water. Balogun and Fagade (2004) reported the LT<sub>50</sub> of 2 d for *B. bassiana* when they were applied to the grasshopper *Zonocerus variegatus*. A wide genetic diversity is observed in different isolates of *B. bassiana*, resulting in isolates differing in their pathogenicity and virulence to different arthropods (Almeida et al. 1997).

The results of investigation on the approach of the trapping system for infection and reinfection of the adults of *R. ferrugineus* with *B. bassiana* showed that the approach was very effective in attracting both sexes of red palm weevils, because of the pheromone and kairomone (Abbass and Mesallam 2013). The insects spent no less than 2 h in the trap, which was enough to

infect the trapped insects directly with a heavy load of spores before they left the trap. The infection with spores of *B. bassiana* were mainly from direct contact application, by walking and climbing the layer of sack-cloth soaked with Broadband. Attachment of a viable spore to the cuticle surface of a susceptible host is an essential stage for growth of the fungus and fungal infection. All insects that were trapped and directly infected were killed after 24h of being trapped in a newly prepared trap (1d old). The directly infected insects of both sexes (female and male) transferred the spores from their bodies to other uninfected adults through their mating behavior. The pathogenic efficacy of the indirect infection was high, as all the indirectly infected insects of both sexes were killed after 3d of mating. The early deaths of the directly infected insects could be due to the high load of *B. bassiana* spores on the body as compared with indirectly infected red palm

weevil. This study also shows the new approach of the trapping system was effective for 13 d. This strategy of controlling the red palm weevil is in match with the previous study finding of Furlong and Pell (1997) when they applied the fungal pathogen *Zoophthora radicans* in strategy for diamondback moth control, based upon a fast entry and slow exit trap containing the female sex pheromone and the pathogen. This strategy kills both adult and larval stages of the diamondback moth. Male moths leave the trap and re-enter the crop, passively vectoring conidia to other susceptible adults and larvae.

The duration of the time required for insects to be attracted to the trap increased as the trap's age increased, and that is due to the reduction in concentration of the pheromone's essences with time (Faleiro 2006, Chakravarthy et al. 2013). In addition, this study demonstrates that the extension in the period of time was required to have the maximum cumulative mortality of the insects. This is due to the fact that the time they were directly or indirectly infected with the fungi increased as the age of the trap increased from the day the Broadband suspension was applied to the sackcloth in the trap. That could be due to degradation of the synergist in the formula, and the reduction of the spore concentration in the trap, or even due to the reduction in the spore viability because of the dry weather at 30–40°C. Hong et al. (1997) reported that temperature and moisture equilibrium is important to fungal spore longevity in storage. Furlong and Pell (1997) found that in laboratory studies some conidia of fungal pathogen *Zoophthora radicans* retained the ability to infect for up to 16 d on the surface of foliage or soil. Higher relative humidity on surfaces resulted in the greater persistence of viable conidia. Therefore, the attachment of the viable spore to the insects became less effective as compared with when it was in a liquid oil suspension.

### Acknowledgments

We would like to thank the Excellence Research Center of Palms and Dates, King Faisal University, for support and technical assistance. We would also thank Dr. Ashraf Amin for statistical assistance.

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Received 1 May 2014; accepted 24 November 2014.

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